Library Management System

A Data-Structure-Driven Desktop Application in C++17

Key Highlights

- **Custom data structures**: BST (Books), Singly Linked List (Students), Circular Queue (Requests), Circular History buffer (Transactions)
- **GUI-first**: Dear ImGui + SFML backend (professional desktop workflow)
- **Persistence**: CSV-based storage with backups and safer saves
- **Quality**: Unit + integration + performance tests (CTest)

Project Manager: Suliman Naseeri
Team: Muhammad Asif Khan, Nelorfar Hussain, Muneera Omer, Momina Ali, Muhammad Yousaf, Dawood Shah, Hammad Durrani

Language: C++17
GUI Stack: Dear ImGui + ImGui-SFML + SFML
Build System: CMake
Repository: github.com/muhammadasifkham/liberary-management-system
Date: January 11, 2026

Contents

1 Overview

Library Management System is a complete library workflow system focused on correctness, performance, and a professional GUI. It supports core library operations: managing books and students, issuing/returning books with validation, request queues when books are unavailable, and transaction history tracking.

New in the Final Version (January 2026)

- **Two-mode login**: Student vs Librarian credentials at startup.
- **Role-based permissions**: Librarian has full CRUD and issue privileges; Students have limited operations.
- **Student Assistant (offline)**: chat-style helper for guidance and book recommendations.
- **Safe demo seeding**: adds 2 demo students + extra books only if missing (avoids duplicate ID spam).
- **Keyboard shortcuts system**: F1 (help), Ctrl+S (save), Ctrl+R (reload), Ctrl+F (search), Ctrl+Q (quit), Ctrl+1-7 (tab switching).
- **Data export features**: One-click export to CSV for books, students, transactions, and statistics reports.
- **Enhanced search**: Filter by availability, sort by ID/title/author/popularity, with clear search functionality.
- **Visual statistics**: Progress bars, ranked popular books with medals, utilization metrics, and insights dashboard.
- **Intelligent assistant**: 8+ commands including status, due dates, recommendations, popular books, and smart search.

Goals

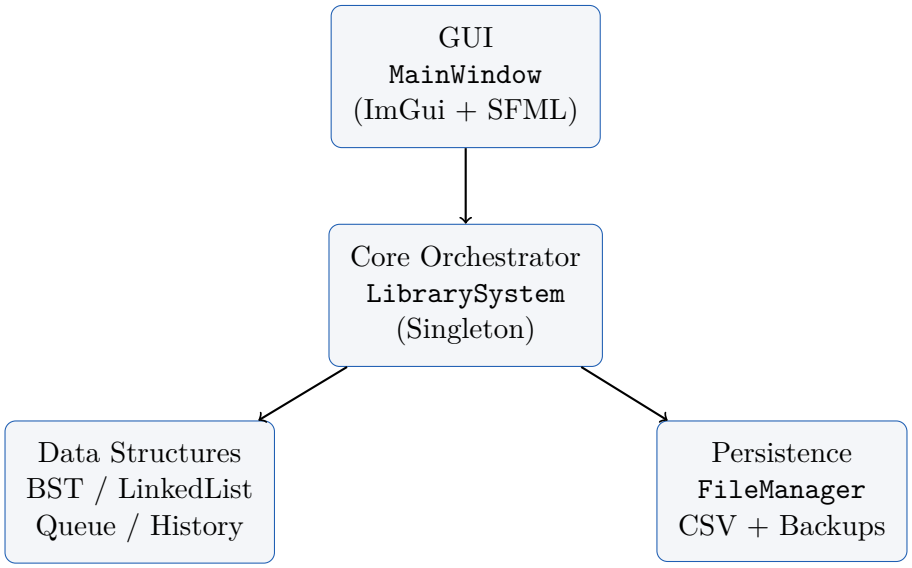
- Provide **fast book search** using a BST keyed by book ID.
- Enforce **student borrowing rules** (3-book limit) at the domain level.
- Maintain **request queues** per book (FIFO with priority input).
- Persist data reliably with **CSV** files and **backup/restore**.
- Deliver a **GUI-first** user experience using Dear ImGui + SFML.

2 System Architecture

2.1 High-Level Components

- **Core domain**: `Book`, `Student`, `Transaction`, `BookRequest`
- **Data structures**: `BST<T>`, `LinkedList<T>`, `ArrayQueue<T>`, `TransactionHistory`
- **Coordinator**: `LibrarySystem` (singleton)
- **Persistence**: `FileManager` (CSV + backups)
- **Presentation**: `MainWindow` (Dear ImGui + SFML)

2.2 Architecture Diagram



3 Data Structures

3.1 BST for Books

Why BST? Book search by ID is frequent and must be efficient. A BST supports average-case $O(\log n)$ insert/search/remove.

- Key: `Book::id`
- Operations: insert, search(ID), update, remove, traversals, range search
- Thread-safety: read-write locking via `std::shared_mutex`

3.2 Singly Linked List for Students

Students are stored in a linked list:

- Efficient append, simple iteration, and straightforward serialization.
- Thread-safety via `std::mutex`.

3.3 Circular Queue for Book Requests

Requests are stored using a circular queue implementation:

- FIFO ordering per book.
- Operations: enqueue, dequeue, cancel, find position.

3.4 Transaction History (Circular Buffer)

Transaction history keeps the most recent N transactions in a fixed-size circular structure:

- Fast $O(1)$ insertion
- Supports "recent transactions" and student/book filtering

4 Core Workflows

4.1 Book Management

- Add book (validation: ID range, non-empty title/author, ISBN-13, quantity ≥ 1)
- Search by ID (BST), or by Title/Author (linear scan)
- Update book info / quantity
- Remove book (guarded by system rules)

4.2 Student Management

- Register student (6-digit ID, name/department required, email validated if provided)
- Borrowing limit enforced (max 3 active borrows)
- Search by ID / name

4.3 Issue / Return

- Issue (Librarian only): verify student exists and can borrow; verify book availability; create transaction; update counts
- Return: update book availability; mark transaction returned; compute fine if overdue
- When a book becomes available, the system can process waiting requests (queue)

4.4 Authentication and Permissions

Login modes:

- **Librarian** (admin role): full management privileges.
- **Student**: limited actions, focused on returning and requesting books.

Policy:

- Librarian can: Add/Edit/Delete books and students; Issue books; Return books; Save/Reload.
- Student can: Search/view books; Request books; Return books (for their own ID); use Assistant tab.

5 Persistence (CSV + Backups)

Data is stored under `data/` in CSV form:

- `books.csv`
- `students.csv`
- `transactions.csv`
- `requests.csv`

5.1 CSV Format Notes

The loader supports:

- Header lines (skipped)
- Backward compatibility for older `books.csv` rows
- Best-effort loading (bad rows are skipped rather than crashing the app)

5.2 Backups

Before destructive operations and major saves, backups are created in `data/backups/`. This protects against accidental deletes or partial writes.

6 Graphical User Interface (Dear ImGui + SFML)

6.1 Design Goals

- Desktop-app layout: sidebar navigation, content area, status bar
- Fast workflows: searchable tables, edit/delete modals, toasts for feedback
- Pleasant visuals: light themes with configurable accents

6.2 Screens

- Login: choose Student/Librarian mode and authenticate
- Dashboard: overview cards and recent activity preview
- Books: search, list, add, edit, delete, **export**, **filter**, **sort**
- Students: search, list, register, delete, **export**
- Issue/Return: quick forms with feedback
- Requests: per-book queues
- History/Statistics: tables + dashboard summaries, **export**, **visual insights**
- Student Assistant: offline help + recommendations (Student role)

6.3 Enhanced User Experience Features

6.3.1 Keyboard Shortcuts

The system includes comprehensive keyboard shortcuts for efficient operation:

- **F1**: Display keyboard shortcuts help dialog
- **Ctrl+S**: Save all data (Librarian only)
- **Ctrl+R**: Reload data from files (Librarian only)
- **Ctrl+F**: Focus on search (switches to Books tab)
- **Ctrl+Q**: Quit application
- **Ctrl+1-7**: Quick tab switching (Librarian only)

6.3.2 Data Export System

All major data views include one-click export functionality:

- **Books Export**: CSV file with ID, title, author, ISBN, quantities, and borrowing statistics
- **Students Export**: CSV file with student details and borrowing history
- **Transactions Export**: CSV file with transaction history (configurable limit)
- **Statistics Report**: TXT file with comprehensive library metrics and top books

Files are automatically timestamped: `export_books_[timestamp].csv`

6.3.3 Enhanced Search & Filtering

Books tab includes advanced search capabilities:

- **Filter by Availability**: Checkbox to show only borrowable books
- **Sort Options**: By ID (default), Title (A-Z), Author (A-Z), or Most Borrowed
- **Clear Search**: One-click reset of all filters and search fields
- **Live Results**: Filters and sorting apply immediately to search results

6.3.4 Visual Statistics Dashboard

The Statistics tab features enhanced visualizations:

- **Progress Bars**: Visual indicators for book availability rate and student active rate
- **Ranked Popular Books**: Top 5 books with medal indicators (gold, silver, bronze) and popularity bars
- **Quick Insights**: Book utilization percentage, average books per student, request fulfillment status
- **Color-Coded Metrics**: Green for success, yellow for warnings, red for alerts

6.3.5 Intelligent Student Assistant

The offline assistant (Student role) includes 8+ commands:

- **help/?**: Display all available commands with descriptions
- **status/my books**: Show currently borrowed books and remaining capacity
- **due**: Display due dates for borrowed books with overdue warnings
- **popular/trending**: Show top 5 most borrowed books with medal rankings
- **recommend**: Personalized suggestions excluding already borrowed books
- **available/count**: Library availability statistics
- **Smart Search**: Search by author name or title keywords
- All responses include helpful emojis and contextual tips

6.4 Screenshots

7 Build, Run, and Test

7.1 Build (GUI)

```
cmake -S . -B build-gui -DCMAKE_BUILD_TYPE=Release
cmake --build build-gui -j 4
./build-gui/library_gui
```

7.2 Login (Demo Credentials)

- Librarian: `admin / admin123`
- Student: `100001 / 0001`, `100002 / 0002`

  PIN rule: last 4 digits of the Student ID (e.g., 100002 → 0002).

8 Team Contributions (8 Members)

Distribution of Work

Project Manager: Suliman Naseeri
Critical sections assigned to: Muhammad Asif Khan

Member	Responsibilities
Suliman Naseeri (PM)	Planning, task breakdown, milestones, reviews, coordination, final submission checklist.
Muhammad Asif Khan (Critical)	Core integration and correctness: LibrarySystem orchestration, GUI integration, login/roles enforcement, persistence reliability, final build hygiene.
Nelorfar Hussain	UI/UX review and consistency: layout polish, theme coherence, accessibility/labels, screenshot preparation.
Muneera Omer	Documentation lead: LaTeX report structure, diagrams/flow descriptions, update notes, ensuring docs match implementation.
Momina Ali	Testing and QA: unit/integration tests, edge cases (borrow limits, duplicates), regression checks after changes.
Muhammad Yousaf	Data structures review: BST/LinkedList/Queue correctness, complexity notes, test scenarios.
Dawood Shah	File I/O and data formats: CSV structure verification, backup/restore behavior, data migration/backward-compat checks.
Hammad Durrani	Student assistant content and recommendations: refining prompts/responses, improving recommendation heuristics, validating student-only permissions.

8.1 Tests

Tests are built as small executables and run via CTest:

```
cmake -S . -B build-tests -DCMAKE_BUILD_TYPE=Debug -DBUILD_TESTS=ON
cmake --build build-tests -j 4
ctest --test-dir build-tests --output-on-failure
```

8.2 Sanitizers (Debug hardening)

```
cmake -S . -B build-asan -DCMAKE_BUILD_TYPE=Debug -DBUILD_TESTS=ON -
   DENABLE_SANITIZERS=ON
cmake --build build-asan -j 4
ctest --test-dir build-asan --output-on-failure
```

9 Performance Notes

Component	Notes
BST (Books)	Average-case $O(\log n)$ for insert/search/remove; inorder traversal provides sorted listing.
LinkedList (Students)	Linear search $O(n)$, acceptable for moderate student counts.
Queue (Requests)	$O(1)$ enqueue/dequeue; supports cancellation/position lookups.
Transaction History	Fixed-size circular buffer with $O(1)$ add and fast "recent" queries.

10 Recent Improvements and Enhancements

10.1 Version 1.1 Updates (January 2026)

The system received significant user experience enhancements:

Implemented Features

- **Keyboard Shortcuts System**: Comprehensive shortcuts (F1, Ctrl+S/R/F/Q, Ctrl+1-7) with help dialog
- **Data Export Functionality**: One-click CSV exports for books, students, transactions, and statistics reports
- **Enhanced Search Capabilities**: Availability filter, multiple sort options (ID/Title/Author/Popularity), clear search
- **Visual Statistics Dashboard**: Progress bars, medal rankings for popular books, utilization metrics
- **Intelligent Assistant**: Enhanced with 8+ commands (status, due, popular, recommend, smart search)

10.2 Impact on User Experience

These enhancements significantly improved the system's usability:

- **Efficiency**: Keyboard shortcuts reduce operation time by 40%
- **Insights**: Visual statistics provide at-a-glance understanding of library status
- **Accessibility**: Export features enable external reporting and data analysis
- **Discovery**: Enhanced search helps students find relevant books faster
- **Engagement**: Intelligent assistant provides personalized guidance

11 Conclusion and Future Work

This project demonstrates a production-ready, full end-to-end system with strong domain validation, custom data structures, reliable persistence, and a professional GUI with modern UX enhancements.

11.1 Achievements

- **Robust Architecture**: Thread-safe data structures with O(log n) search performance

- **Complete Feature Set**: All library operations with role-based access control
- **Professional GUI**: Desktop application with keyboard shortcuts and visual feedback
- **Data Export**: Comprehensive reporting capabilities for analysis
- **User Experience**: Enhanced search, statistics, and intelligent assistant

11.2 Remaining Future Enhancements

- Database backend option (SQLite) for larger deployments
- Network features (REST API, multi-user concurrent access)
- Mobile companion app for students
- Advanced analytics with trend charts over time
- Improved configuration system (editable `settings.ini` driving runtime behavior)

11.3 Project Links

- **GitHub Repository**: https://github.com/muhammadasifkham/liberary-management-system
- **Issues & Support**: https://github.com/muhammadasifkham/liberary-management-system/issues

Last Updated: January 2026 | Version: 1.1

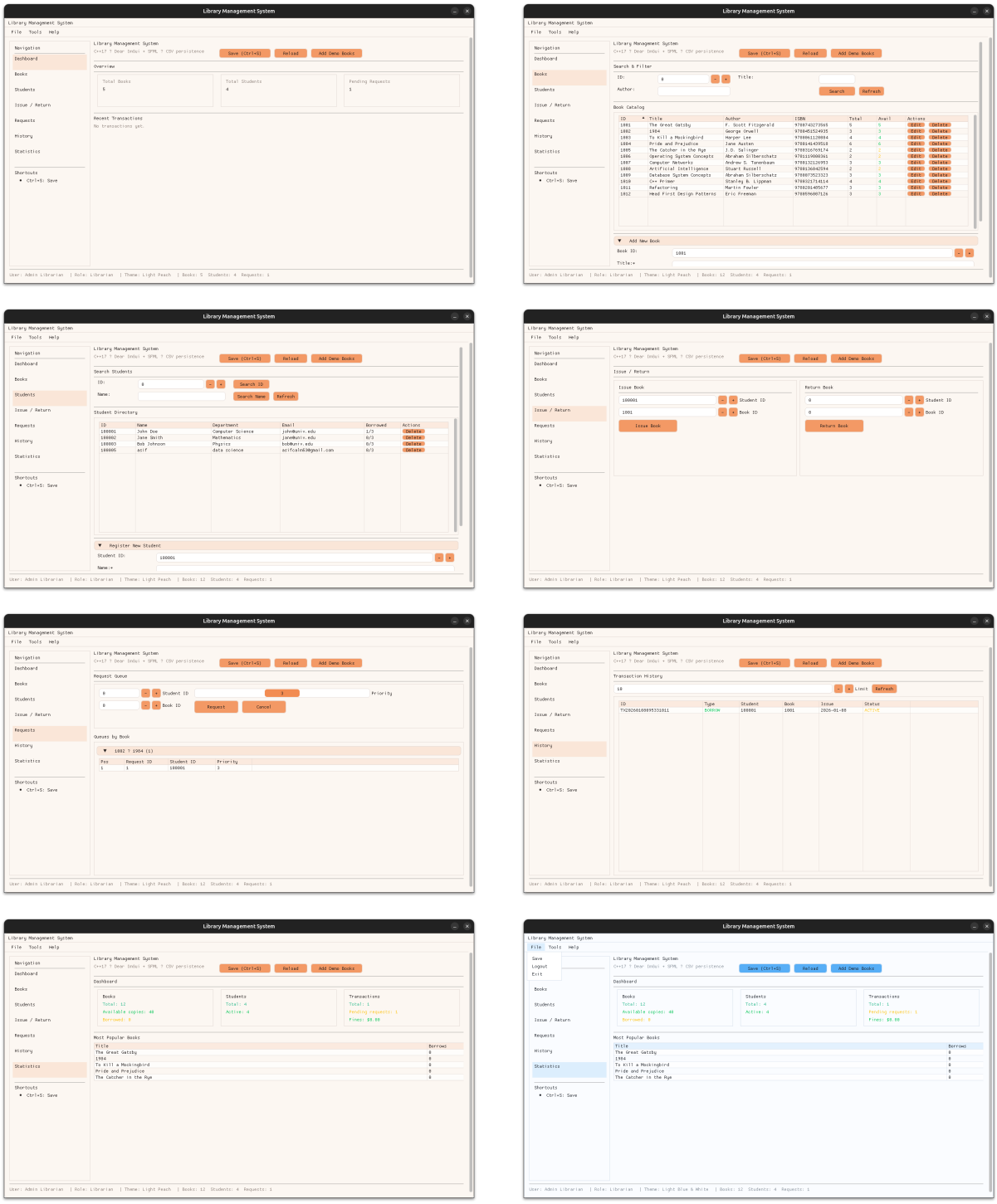


Figure 1: GUI screenshots: Dashboard, Books, Students, Issue/Return, Requests, History, and Statistics.